Max Tikhomirov

AI-first Senior Software Engineer

Cross-platform systems · AI agent tooling · Products shipped solo

C++ Rust Elixir Swift

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Remote

PROFILE

I build systems and ship complete products end to end.

Direct, precise, proactive, autonomous.

Senior engineer at a US video management company whose clients include SpaceX, Tesla and Bank of America, where I designed and built cross-platform plugin isolation, shared-memory video transport and an AI-assisted engineering environment. Independently, I build products and developer tools: GymFlow, a native SwiftUI iOS app, and Hive, an open-source Rust runtime for persistent Codex and Claude agents. Earlier: aerospace ground-control software and a patented code-generation system. I started programming at 34 with no technical background.

DISTINCTIONS

Patent	RU2780080: granted for a code-generation system I invented, which became core development infrastructure at an aerospace company.
Certification	World top 10 in two C++ certifications, two years after writing my first line of code.

EXPERIENCE

US Video Management Systems Provider

Jun 2023 – Present

Senior Software Engineer · Remote

Enterprise video management platform; clients include SpaceX, Tesla and Bank of America.

- Designed and built analytics-plugin isolation across Linux, Windows and macOS: plugins moved out of the media server into dedicated, sandboxed, resource-limited hosts with supervision, crash reporting and automatic recovery.
- Introduced shared-memory video delivery: frames are prepared once and shared read-only across plugin processes, with explicit ownership and bounded memory.
- Designed asynchronous private IPC carrying the full plugin SDK, so existing plugins run unchanged.
- Built an AI-assisted engineering environment that automates evidence collection, worktrees, code review, CI investigation and testing; extended it into automated analytics QA, using QA workflows over a distributed network of remote machines.
- Earlier: built the REST/WebSocket analytics API that opened the plugin SDK beyond C++, and proposed development-process changes adopted company-wide.

GymFlow

2026 – Present

Founder & Engineer · gymflowai.fit · App Store

Designed, built and launched GymFlow, an iOS workout app with AI-generated workouts.

- Owned the full product: native SwiftUI client, Elixir/Phoenix backend, PostgreSQL, RevenueCat subscriptions, OpenAI integration and iOS Live Activities.
- Took it from spec to App Store approval.
- Launched on React Native, then rewrote the client from scratch in SwiftUI: client size improved **3×**, with a native look and feel.
- Open-sourced four packages extracted from the codebase: **idempotency_kit**, **mobile_id_token** and **revenue_cat** (Hex), and **idempotent-fetch** (npm).

EXPERIENCE CONTINUED

UWCA

Jul 2020 – Jun 2023

C++/Scala Developer · Remote

Ground-control software for drones and manned aircraft.

- Architected and built the ground-control GUI from scratch in C++ (Qt Widgets/QML) and shipped it to production deployments.
- Designed a JSON-spec-to-C++ code-generation system with a transparent network layer, granted as patent RU2780080. Prototyped in about **1,000 lines** of Haskell, it replaced **tens of thousands of lines** of hand-written C++ and became core development infrastructure.
- Debugged a live avionics integration from video recordings of logs on an air-gapped onboard machine, and found and fixed errors in the protocol specification.
- Built ScalaJS map-editor tooling and Java gRPC clients for hardware devices.
- Cut Scala environment setup from a **2–4 hour** assisted process to **30 minutes** of self-service.
- Mentored an intern with no prior experience into a full-time role and later promotion; interviewed C++ candidates.

Tensor (SBIS)

Sep 2018 – Nov 2019

C++ Developer

Server-side development for a large enterprise-management platform, plus desktop products for Windows, macOS and Linux, with supporting Python and Bash tooling.

PROJECTS

Hive

github.com/metacircu1ar/hive

Open-source runtime that turns Codex and Claude into a persistent team of agents: an orchestrator plus a shared brain with semantic search.

hive_skills

github.com/metacircu1ar/hive_skills

Playbooks that make AI coding agents plan, build, review and audit like a disciplined senior team. The review protocol is model-checked in TLA+.

atomic-fs

github.com/metacircu1ar/atomic-fs

Crash-safe file writes for Rust. The storage foundation under Hive.

blazing-radix

github.com/metacircu1ar/blazing-radix

The fastest radix sort in Rust: beats every leading crate on every benchmark. Zero dependencies.

More open-source skill libraries and Rust, Elixir and JavaScript libraries: github.com/metacircu1ar

SKILLS

Languages	C++ · Rust · Swift · Elixir · Haskell · Scala · Java · Python · JavaScript · Bash
Systems	Linux, Windows and macOS · process isolation and sandboxing · IPC · shared memory · supervision and crash recovery · Qt Widgets/QML
AI tooling	agent runtimes (Codex, Claude) · AI-assisted code review · CI investigation and testing · automated QA · semantic search · TLA+
Product	SwiftUI · React Native · Elixir/Phoenix · PostgreSQL · RevenueCat · OpenAI integration · iOS Live Activities · REST/WebSocket · gRPC